

Problem Solving Practice

HW Problem #5

CSE4152
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Who is the Richest?

There are N people. Each person has a certain amount of wealth, but initially, you do not know how much money each person has. The wealth of every person is at most 10^9 . You cannot directly check a person's wealth. Instead, you may ask the following type of question any number of times:

“Does the i -th person have at least x won?”

Each question is answered with either **Yes** or **No**.

Using these questions, you must determine one person with the greatest wealth among the N people, as well as the exact amount of wealth that person has. If multiple people share the greatest amount of wealth, you may output any one of them. Find a way to determine the answer using the **minimum possible number of questions**.

Who is the Richest?

Example:

Wealth of person 1: 20 won

Wealth of person 2: 5 won

Wealth of person 3: 10 won

Example Questions

Does person 1 have at least 5 won? Yes

Does person 1 have at least 21 won? No

Does person 1 have at least 20 won? Yes

Does person 2 have at least 20 won? No

Does person 3 have at least 20 won? No

Answer:

Person 1, 20 won

Chasing Light

There is a robot moving on an infinite integer grid. The robot can recharge its energy using sunlight, and it travels across the grid to receive as much light as possible. Its goal is to gain as much energy as possible during the given time period.

Initially, the robot is located at $(0,0)$. Every second, it may choose one of the following two actions:

- **Move:** Move by 1 unit to one of the four adjacent cells: up, down, left, or right. The robot uses gasoline for movement, so moving does not decrease its stored energy.
- **Charge:** Stay at the current position. If light is shining on the current position at that time, the robot gains 1 unit of energy.

Light appears a total of N times. The i -th light shines at coordinate (X_i, Y_i) during the time interval $[S_i, E_i]$. No two lights shine on the same coordinate at the same time. The robot knows in advance when and where every light will appear. Therefore, it can freely decide which lights to move toward, when to move, and when to stay still and charge.

Determine the maximum amount of energy the robot can gain if it chooses its movements and charging actions optimally.

Chasing Light

Example:

$N = 4$

0 0 0 4

4 0 5 8

0 3 5 9

1 3 10 13

Answer:

9

